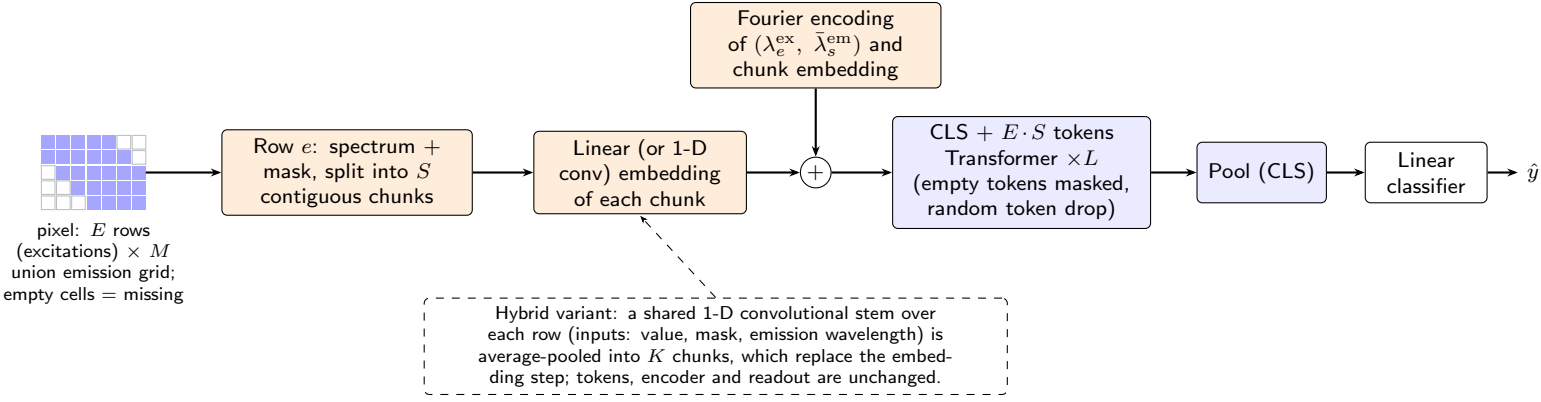

Fourier encoding of $(\lambda_e^{\mathrm{ex}},\ \bar{\lambda}_s^{\mathrm{em}})$ and chunk embedding
pixel: E rows (excitations) $\times$ M union emission grid; empty cells = missing
Row e: spectrum + mask, split into S contiguous chunks
Linear (or 1-D conv) embedding of each chunk
CLS + $E \cdot S$ tokens Transformer $\times L$ (empty tokens masked, random token drop)
Pool (CLS)
Linear classifier
$\hat{y}$
Hybrid variant: a shared 1-D convolutional stem over each row (inputs: value, mask, emission wavelength) is average-pooled into K chunks, which replace the embedding step; tokens, encoder and readout are unchanged.